

Semantic Engines: An Introduction to Mind Design

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I Cognitive Science

"Reasoning is but reckoning," said Hobbes (1651, ch. V), in the earliest expression of the computational view of thought. Three centuries later, with the development of electronic "computers," his idea finally began to catch on; and now, in three decades, it has become the single most important theoretical hypothesis in psychology (and several allied disciplines), and also the basis of an exciting new research field, called "artificial intelligence." Recently, the expression *cognitive science* has been introduced to cover all these varied enterprises, in recognition of their common conceptual foundation. This term, therefore, does not apply to every scientific theory of cognition, but only to those sharing a certain broad outlook – which is sometimes called the "information processing" or "symbol manipulation" approach. Perhaps, at last, Hobbes's philosophical insight has found its home in a proper scientific paradigm (Kuhn, 1970). [...]

Often, discussion of cognitive science focuses on *artificial intelligence* – "AI," among friends – because it amounts to a kind of distilled essence of cognitive science. But again, it is important to realize that "AI" (like "cognitive science") is more specific in its meaning than the words themselves might suggest. Crudely, we can put the point in terms of different technologies: a project at IBM to

wire and program an intelligent robot would probably be AI, whereas a project at DuPont to brew and mold a synthetic-organic android probably would not. But this can be misleading; the crucial issue is not protoplasm versus semiconductor ("wetware" versus "hardware"), but rather whether the product is designed and specified in terms of a computational structure. If it is, then a working model could probably be manufactured much more easily by means of electronics and programming; and that's the *only* relevance of the technology. Indeed, the guiding inspiration of cognitive science is that, at a suitable level of abstraction, a theory of "natural" intelligence should have the same basic form as the theories that explain sophisticated computer systems. It is this idea which makes *artificial* intelligence seem not only possible, but also a central and pure form of *psychological* research.

A better perspective on all the excitement can be gained by asking why it took three hundred years for Hobbes's original proposal to be appreciated. Mainly, three famous philosophical dilemmas stood in the way: (i) the metaphysical problem of mind interacting with matter; (ii) the theoretical problem of explaining the relevance of meanings, without appealing to a question-begging homunculus; and (iii) the methodological issue over the empirical testability (and, hence, respectability) of "mentalist" explanations. The computational idea can be seen as slicing through all three dilemmas at a stroke; and this is what gives it, I think, the bulk of its tremendous gut-level appeal.

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Descartes, a contemporary of Hobbes, gave the mind/matter problem its modern form in his doctrine of metaphysical *dualism*. Mind and body, he said, are two entirely different *kinds* of substance: the one can have (as distinguishing characteristics) various thoughts and feelings, whereas the other can have shapes, motions, and the causal interactions described by physical laws (and not vice versa). Intuitively, this is much more appealing than *materialism* (the main alternative to dualism), according to which everything, including minds, is really just matter, in one form or another. Not only are we reluctant to ascribe thought and feeling to "mere" matter, but we also find it very hard to ascribe shape and location to minds or ideas. There is, however, one basic problem, which no dualist has ever really solved: how can mind and body *interact*? On the one hand, they certainly *seem* to interact, as when a mental decision leads to a physical action, or when a physical stimulus leads to a mental perception; indeed, it's not clear how perception and action could be possible at all without mind/body interaction. On the other hand, however, physical laws are supposed to describe all motions of all bodies *completely* in terms of their interactions with one another.¹ In other words, physics leaves no room for causal intervention by the mental; hence the price of mind/body interaction is violation of the laws of physics – a price that few philosophers (or scientists) are willing to pay.

Thought itself (quite apart from matter) is not static and not random: it progresses and develops in ways that obey (at least much of the time) various rules of inference and reason. Superficially, this suggests an analogy with material particles obeying the laws of physics. But the analogy breaks down at a crucial point: particles have neither choice nor difficulty in "obeying" physics – it happens infallibly and automatically. People, on the other hand, often have to work to be reasonable; following the rules of reason is hardly infallible and can be very difficult. But this means there cannot be an explanatory dynamics of thought, which is at all comparable to physical dynamic theories; the respective roles of rules and laws in the two cases are deeply different. In particular, since correct application of the rules of reason to particular thoughts depends on what those thoughts *mean*, it seems that there must be some active rule-applier, which (or: who) *understands* the thoughts (and the rules), and which applies the rules to the thoughts as well as it can. If the activity of this rule-applier, following the rules of

reason, is to explain the rationality of our thought processes, then it must be regarded as a complete little person – or *homunculus* (in Latin) – inside the head, directing the thoughts like a traffic cop. The trouble is: a theory that invokes an homunculus to *explain* thinking has begged its own question, because the homunculus itself has to think, and *that* thinking has *not* been explained.

Finally, there is the question of how the psychology of thought could ever be properly scientific. Thoughts, it seems, cannot be observed; and the difficulty is not that, like electrons or distant galaxies, they are too small or too far away. Rather, they are somehow essentially subjective – we don't even know what it would be like to observe (or measure) them objectively. So all that science has to go on, even in principle, is objective *behavior* (which might include internal "physiological behavior"); hence, thoughts can enter the picture only as inferred or hypothesized intermediates. Unfortunately, in any given case, invoking beliefs and desires to explain an action is just too easy. Whatever an organism does, one can always trump up a million different ideas and motives that would explain it. If there can be, in principle, no independent, empirical check on which of these hypotheses is the right one, it seems scientifically disreputable to accept any of them, even tentatively. This roughly, is the stance taken by *behaviorism*. The other half of the story, however, is that explaining behavior *without* invoking intervening mental processes is just too hard. In general, the subtle regularities, connection, and nonrandom variations in real-life behavior cannot so much as be *described* in nonmentalistic (pure behaviorist) terms, let alone *explained* – hence, the overwhelming mechanicalness and stupidity of all the phenomena that the behaviorists were ever really able to account for.² The upshot is another philosophical standoff, with mentalists and behaviorists gleefully accusing one another of scientific bankruptcy.

Cognitive scientists can be materialists (nondualists) and mentalists (nonbehaviorists) at the same time; and they can offer explanations in terms of meaning and rule-following, without presupposing any unexplained homunculus. It all depends on a marvellously rich analogy with computers – the outlines of which we can see with a quick look at everybody's favorite example: a chess-playing machine. It would be very awkward and peculiar to start assigning geometrical shapes and locations to the internal program routines and operations

(decision processes and data structures, say) of such a system; yet we are quite confident it has no immaterial soul. These same decisions clearly cause physical behavior (e.g., in teletypewriters or TV screens), yet no one is worried that the laws of physics are being violated. When the machine plays, it follows rules in at least two senses: it always abides by the rules of the game, and it employs various reasonable rules of thumb to select plausible moves. Though these rules are in no way laws of nature, the machine's behavior is explained (in part) by citing them – and yet, no unexplained “compunculus” is presupposed. Finally, this explanation will necessarily invoke the system's internal reasoning processes; yet it is far from easy to figure out (or design) processes that will consistently lead to the observed (or desired) behavioral responses. Moreover, for any given machine, on any given occasion, there seems to be a determinate right answer about which reasonings it in fact went through.

That may have been a bit swift, but still, *what an inspiration!* If there are no philosophical dilemmas about chess-playing computers, then why should there be any about chess-playing people – or, indeed, about human intelligence in any other form? To put it coldly: why not suppose that people *just are* computers (and send philosophy packing)? Well... nothing very interesting is ever that simple. Various questions come up, among the first of which is: What exactly is being proposed, anyway?

II Formal Systems

To start at the beginning, we must first say a little bit more carefully what a computer is. It is an automatic formal system. To see what this means, we first consider what a formal system is, and then what it is to automate one.

A *formal system* is like a game in which tokens are manipulated according to rules, in order to see what configurations can be obtained. Basically, to define such a game, three things have to be specified:

- 1 what the tokens are;
- 2 what the starting position is; and
- 3 what moves are allowed in any given position.

Implicit in (2) and (3) is a specification of what positions are possible (for instance, what the board is, if it's a board game). Also, there is sometimes a specified *goal* position, which the player (or each

player) is trying to achieve – such as a “winning position.”

For example, there is a familiar solitaire game in which the tokens are pegs, arranged as follows in the starting position:



The solid dots are the tokens (pegs), and the circle in the middle is an empty space, into which a token could fit. The only move allowed by the rules is jumping one token over an adjacent one into an empty space, and then removing the token jumped over. This game has a goal: to perform such jumps until only one token remains, and it is in the center space.

Three points should be noticed about this (simple-minded) formal system. First, it is entirely *self-contained*. Only its own tokens, positions, and moves make any difference to it, and these only insofar as they matter to the application of the rules. In other words, the “outside world” (the weather, the state of the economy, whether the building is on fire, and so on) makes no difference whatsoever in the game. And, further, any aspects of the tokens and positions themselves which are irrelevant to determining which moves are legal – e.g., (in this game) color, size, weight, market value – are equally outside the game. Politics and courtship, by contrast, are not at all self-contained (even though they are sometimes called games) because just about anything could be relevant in some situation or other. Second, every relevant feature of the game is *perfectly definite*; that is, barring outright mistakes or breakdowns, there are no ambiguities, approximations, or “judgment calls” in determining what the position is, or whether a certain move is legal. For each peg and slot, that peg is either definitely (obviously and 100 percent) in that slot, or definitely (obviously and 100 percent) not in that slot – there are no in-between or borderline cases. Third, the moves are *finitely checkable*, in the sense that for each position and each candidate move, only a finite number of things has to be checked to see whether that move would be legal in that position. This is pretty trivial for our example, but it's nontrivial and very

It is sometimes said (with a grand air) that *any* analog device can be digitally simulated to *any* desired precision; but this is grossly misleading. Digital simulation is possible only when all the operative relationships in the analog system can be described in a relatively compact and precise way – e.g., not too many equations with not too many variables. The architectural simulation, for example, depends on assuming that all the light comes from the same direction and travels in a straight line until it's blocked by one of the few specified opaque surfaces. Without this simple structure to work from, the simulation could not get off the ground. Thus there is no comparable general description of the physiology of laboratory rats pumped full of odd chemicals – there are billions of potentially relevant subtle interactions and responses in a complex organic system, and the scope for combinatorial explosion is essentially indescribable. Hence, digital simulation is out of the question, except in very special cases where it is known in advance that only a certain few variables and relationships are relevant. It might seem that, in principle, simulation must be possible anyway, because rats are made of atoms, and each atom obeys known equations. But such a principle is utterly out to lunch. A single large molecule may have so many interdependent degrees of freedom, that no computer yet built could simulate it reliably in reasonable time; and one rat contains more molecules than there would be minicomputers if the entire volume of the earth were packed solid with them.

The real theoretical advantage of digital systems lies in quite another direction, and depends specifically on their digital nature. Consider two conventions for keeping track of money in a poker game. Each uses different colors for different denominations: blue, red, and white for a hundred, ten, and one, respectively. But in one system, the unit of each denomination is a colored plastic disk (i.e., a poker chip), whereas in the other system it is a tablespoon of colored sand. The latter arrangement does have some merits – particularly the fact that *fractional* bets are possible – using less than a full tablespoon of white sand. But the chip system has one overwhelming advantage: all the bets are *exact*. By contrast, in measuring volumes of sand, there is always some small error, no matter how careful you are. Using the chips, and a very modest degree of care, it is possible to wager *exactly* 13 units (plus or minus zero); but with the sand this is impossible, even given the finest instruments in

the world. The difference is that the chip system is digital; each token is perfectly definite, and there is no need for judgment calls.

The more complex and interdependent a system becomes, the more vulnerable it is to errors that propagate and get out of control. A small error in one component, affecting a more sensitive component, can result in a larger error going to the next component, and so on. We can see a rudimentary form of this, even in the poker example. Suppose the measurement error on sand volumes is ± 2 percent; then it would be perverse to try to bet, say, 613 units, because the measurement error on the six blue tablespoons is worth four times as much (on the average) as all three white tablespoons – the latter simply drop out as irrelevant. There are ways to control such errors, of course, but only to a certain extent, and they tend to get expensive. The perfect definiteness of digital tokens, though confining in some cases, pays off in the elimination of this sort of error; thus, though you cannot bet a fraction of a unit with poker chips, there is no problem in betting 613 units – since six blue chips can be counted exactly, the white chips remain perfectly significant. This advantage is progressively more important for larger and more complicated systems – hence the contemporary predominance of digital methods.

VI Semantics

Formal systems (and computers) can be more than mere games, because their tokens can have interpretations that relate them to the outside world. This is the domain of semantics and pragmatics.

Sometimes we say that the tokens in a certain formal system *mean* something – that is, they are “signs,” or “symbols,” or “expressions” which “stand for,” or “represent,” or “say” something. Such relations connect the tokens to the outside world (what they are “about”), making it possible to use them for purposes like record-keeping, communication, calculation, and so on. A regular, systematic specification of what all the tokens of a system mean is called an *interpretation*; and the general theory of interpretations and meanings is called *semantics*. Accordingly, what any token means or says, and hence also whether it is true or false, and so on, are all *semantic properties* of that token.

Semantic properties are not formal properties. A formal system as such is completely self-contained

and, viewed in that way, is just a meaningless game. In strictly formal terms, interpretation and meaning are entirely beside the point – they are extraneous “add ons” that are formally irrelevant. (When discussing a system that is to be interpreted, we call its purely formal characteristics and structure its *syntax*; “syntactical” is just another word for “formal,” but it is generally used only when a contrast to “semantic” is relevant.)

So, formal tokens can lead two lives: *syntactical* (formal) *lives*, in which they are meaningless markers, moved according to the rules of some self-contained game; and (if the system is interpreted) *semantic lives*, in which they have meanings and significant relations to the outside world. The story of how these two lives get together is the foundation of modern mathematics and logic; and it is also the philosophical inspiration of cognitive science. We review the mathematical/logical case first, however, because it is better understood and there are fewer complications.

Consider a formal system (like the algebra game, discussed above in section 2) in which the positions are just sets of tokens, and the legal moves are just to add more tokens to the current position (depending, of course, on what is already in it). And suppose this system is interpreted in such a way that each of its (well-formed, complete) tokens “asserts” something – that is, each token (according to the meaning assigned to it by the interpretation) expresses some claim about the world. Then, depending on what the claim is and what the world is like, each of these tokens will be either true or false (relative to that interpretation). Now, the *rules* of such a system will have the semantic property of being *truth-preserving* if and only if they meet the following condition: for any position which already contains *only* true tokens, any other token which can be added to that position (according to the rules) will also be true. In other words, if you start with tokens which are all true (under the interpretation), and if you obey the (purely formal) rules, then you can be sure there are no false tokens in any position you ever reach.

The rules of standard logical and mathematical systems, relative to their standard interpretations, are all truth-preserving; and, of course, the tokens in their starting positions (i.e., their axioms) are all true. Therefore, any token in any legal position of one of these systems is guaranteed also to be true! That is why we know in advance that their *theo-*

rems (which are defined in purely syntactical/formal terms) are all *true* (which is a semantic property). Or, what comes to the same thing, in order to establish the semantic truth of a token in such a system, it suffices merely to prove it formally (play the game). This is how the “two lives” of the tokens get together; and it is the basic idea behind the formalization of modern logic and mathematics. In effect, given an interpreted formal system with true axioms and truth-preserving rules, if you take care of the syntax, *the semantics will take care of itself*.⁷

Most mathematical and logical systems are only partially interpreted, in the sense that some of their atomic tokens are left as variables, whose exact interpretation is to be specified by the user (in certain allowable ways). For example, in ordinary algebra, you can specify what the letters (variables) stand for in any way you want – so long as they stand for numbers. The important thing is that the rules are still truth-preserving, no matter what specific interpretation is given to the variables. For example, if a user knows, relative to some specific interpretation of the variables (as values in a physics problem, say) that the token

$$\frac{a-b}{c} + c = \frac{d-b}{c} + \frac{e(1+e)-e}{c}$$

is true, then he or she can apply the purely *formal* rules for multiplying through, deleting parentheses, collecting terms, etc., and be assured that the token

$$a + c^2 = d + c^2$$

is also *true* (relative to the same interpretation). This is a vivid example of how useful it can be to have the semantics “take care of itself,” if only one plays the syntactical game correctly.

An even more vivid example is computers, because precisely what an automatic formal system can do is “take care of the syntax” – i.e., play by the rules. The machine does not have to pay any attention to the interpretation or meaning of any of its tokens. It just chugs along purely formally; and if the starting position, rules, and interpretation happen to be of the sort we have been discussing, then it will automatically produce only truths. Given an appropriate formal system and interpretation, the semantics takes care of itself. This, fundamentally, is why computers can be so useful for calculations and the like – why they can be *computers* and not just electronic toys.

An automatic formal system with an interpretation such that the semantics will take care of itself is what Daniel Dennett (1981) calls a *semantic engine*. The discovery that semantic engines are possible – that with the right kind of formal system and interpretation, a machine can handle meanings – is the basic inspiration of cognitive science and artificial intelligence. Needless to say, however, mathematics and logic constitute a very narrow and specialized sample of general intelligence. People are both less and much more than automatic truth-preservers. Consequently, our discussion of interpretation needs to be expanded and generalized considerably; unfortunately, the issues will get messier and murkier as we go.

VII Interpretation and Truth

Interpretation is especially straightforward in the special case of logic and mathematics because, in the final analysis, the only semantic property that matters is truth. Hence, it suffices if the theorems are guaranteed true, given the truth of the axioms (and special assumptions, if any) – that is, it suffices if the rules are truth-preserving. We shall see in the next section that there are many other cases where truth is far from all that matters in judging interpretations. But it is worth staying with the special case a little longer, to ask *why* truth has the importance it does; then it will be clearer why other considerations are important in other cases.

Imagine finding an automatic formal system that uses ordinary numerical tokens, and generates “theorems” (i.e., outputs) like the following:

$$\begin{array}{lll} = 5 = 1 \div & = \div 3 - 1 - 8 & + 1 - 5940 \\ 71 \div 92 \times & = 61 = 040 & 84 - 1 \times 5 = \end{array}$$

Formally, of course, there is nothing wrong with these tokens; we can imagine any number of (strange and boring) games in which they would be perfectly legal moves. *Semantically*, on the other hand, they look like nonsensical, random gibberish – “arithmetic salad.” That is, it seems impossible to construe them as expressing claims about the relationships among numbers (e.g., equations). But hold on: this reaction depends on a tacit adoption of the familiar Arabic interpretation of what the numerals and signs mean (the digit “1” stands for the number one, the “+” sign stands for addition, and so on). Formally,

however, these numerals and signs are just neutral marks (tokens), and many other (unfamiliar) interpretations are possible (as if the outputs were in a code). Suppose, for instance, we construed the atomic tokens according to the following non-customary scheme (using “ $\Rightarrow$ ” to abbreviate “stands for”):

$$\begin{array}{lll} \text{“1”} \Rightarrow \text{equals} & \text{“6”} \Rightarrow \text{zero} & \text{“+”} \Rightarrow \text{five} \\ \text{“2”} \Rightarrow \text{plus} & \text{“7”} \Rightarrow \text{one} & \text{“-”} \Rightarrow \text{six} \\ \text{“3”} \Rightarrow \text{minus} & \text{“8”} \Rightarrow \text{two} & \text{“} \times \text{”} \Rightarrow \text{seven} \\ \text{“4”} \Rightarrow \text{times} & \text{“9”} \Rightarrow \text{three} & \text{“} \div \text{”} \Rightarrow \text{eight} \\ \text{“5”} \Rightarrow \text{div. by} & \text{“0”} \Rightarrow \text{four} & \text{“=”} \Rightarrow \text{nine} \end{array}$$

Then, with this table, we could translate the system’s outputs back into the familiar notation as

$$\begin{array}{lll} 9 \div 9 = 8 & 98 - 6 = 62 & 5 = 6 \div 3 \times 4 \\ 1 = 83 + 7 & 90 = 94 \times 40 & 2 \times 6 = 7 \div 9 \end{array}$$

Superficially, these look more like equations, and hence not nearly as random or crazy as the raw (untranslated) outputs. Unfortunately, they’re all false – *wildly* false. In fact, on closer inspection, the digits look just as random as before; we still have arithmetic salad, only disguised in regular equation format.

But there are over a trillion possible ways to interpret these fifteen atomic tokens (even sticking to permutations of the ordinary one). Here is just one more possibility, together with the translations it would yield for the original outputs:

$$\begin{array}{lll} \text{“1”} \Rightarrow \text{equals} & \text{“6”} \Rightarrow \text{zero} & \text{“+”} \Rightarrow \text{five} \\ \text{“2”} \Rightarrow \text{div. by} & \text{“7”} \Rightarrow \text{nine} & \text{“-”} \Rightarrow \text{four} \\ \text{“3”} \Rightarrow \text{times} & \text{“8”} \Rightarrow \text{eight} & \text{“} \times \text{”} \Rightarrow \text{three} \\ \text{“4”} \Rightarrow \text{minus} & \text{“9”} \Rightarrow \text{seven} & \text{“} \div \text{”} \Rightarrow \text{two} \\ \text{“5”} \Rightarrow \text{plus} & \text{“0”} \Rightarrow \text{six} & \text{“=”} \Rightarrow \text{one} \end{array}$$

$$\begin{array}{lll} 1 + 1 = 2 & 12 \times 4 = 48 & 5 = 4 + 7 - 6 \\ 9 = 27 \div 3 & 10 = 16 - 6 & 8 - 4 = 3 + 1 \end{array}$$

What a difference! These not only look like equations, they *are* equations – they are *true*. And, intuitively, that strongly inclines us to prefer this interpretation – to think it “better” or “righter” than the ones which yield random nonsense as readings. This intuition has several theoretical considerations behind it.

In the first place, formal tokens in themselves never intrinsically favor one interpretation scheme over any other – from their point of view, all interpretations are equally extraneous and arbitrary. So if some particular interpretation is to be

adopted, over all the other possibilities, there must be something distinctive about it, which makes it stand out in the crowd. But, as our example suggests, an interpretation that renders a system's theorems as truths is a rare and special discovery (the alternatives yield mostly gibberish). Hence, such an interpretation is distinctive.

Second, if we regard an interpretation as *relating* or *connecting* a formal system to the outside world (i.e., to whatever its tokens are "about"), then the distinctiveness of the preferred interpretation should lie in that relation. But if a system's theorems are all true under an interpretation, then there is, in fact, a relation strong enough to support inferences about the world from the theorems, and about the theorems from (knowledge of) the world. Thus we can discover new facts about numbers, using an interpreted formalization of arithmetic; likewise, the above-noted (practical) utility of the semantics "taking care of itself" depends on the *truth*-preservingness of the rules. In the other direction, we (sometimes) can predict theorems, given what we know about numbers. For instance, if our example system produced the theorem fragment "7401...", we would be hard pressed to guess its completion, from its form alone. The first interpretation scheme (which gives " $1 \times 4 = \dots$ ") is no help either, because its "translations" come out randomly false. But knowing that the second scheme gives " $9 - 6 = \dots$ ", and that the theorem will be true, makes the prediction easy: the completion will be whichever token means "three" (namely, " $\times$ "). The reliability of such inferences (both ways) indicates that *this* relation between the tokens on the one hand and the numbers and operations on the other is *not* arbitrary – i.e., this interpretation somehow genuinely connects them.

Finally, and most important, interpreting is tantamount to "making sense of"; hence, if the system doesn't end by making sense (but, rather, makes nonsense), then the interpretation attempt has failed. Arithmetic salad does not make sense (whether clothed in the outer form of equations, or not); and that, primarily, is why random interpretation schemes don't really give *interpretations* at all. In the context of arithmetic, true equations make eminently good sense – hence the preferability of our second scheme. (An occasional lapse could be understood as meaningful though false; but constant, random falsehood is, in effect, impossible – because the tokens cease to make any sense at all.) So truth matters to interpreta-

tions not only because it provides a nonarbitrary choice among candidate schemes and because this choice reflects some relation between the system and what it is (interpreted to be) "about," but also, and most of all, because wild falsehood amounts to *nonsense* – the antithesis of meaning.

VIII Interpretation and Pragmatics

In most activities involving meaningful tokens (other than logic and mathematics) truth is far from the only semantic property that matters. Take ordinary conversation, for example. In the first place, many speech acts – such as questions, commands, expletives, and even most banter and quips – are neither true nor false; so some other kind of appropriateness must be what matters. And even in the case of statements (which typically *are* true or false), much else is important besides just whether they are true. For instance, in conversation it is also important (usually) to stick to the topic, be polite, say only what is worth saying (not obvious, inane, redundant, etc.) – and, in general, to avoid making an ass of oneself. It is not clear to what extent such conditions should be called "semantic" (some authors prefer to call them "pragmatic"), but they are all relevant to the acceptability of an interpretation as soon as you get away from special cases like logic and mathematics.

In these cases we are driven back to the more general but also somewhat fuzzier notion of "making sense" as our criterion for the adequacy of interpretations. There is, to my knowledge, no satisfactory philosophical account of what it is to make sense; indeed, it is questionable whether a precise, explicit definition is even possible. Still, there is much to be said, and philosophers have worked out a variety of rough characterizations and prerequisites (sometimes referred to by the peculiar misnomer "principles of charity"). The truth of simple, uncontroversial declarative outputs (i.e., tokens interpreted as such) is, of course, one rough prerequisite on making sense in general; and a few philosophers (notably, Davidson, 1973) are inclined to stick with that. But most would agree there are other considerations.

The most widely discussed, *rationality*, is loosely analogous to truth-preservation in mathematical systems. The idea is that "obvious consequences" of tokens in the current position should be relatively easy to evoke from the system